

8-4b Estimating Uncollectibles

The allowance method requires an estimate of uncollectible accounts at the end of the period. This estimate of current expected credit losses is normally based on past experience, industry averages, current economic conditions, and forecasts of the future.

The two methods used to estimate uncollectible accounts are as follows:

- Percent of sales method.
- Analysis of receivables method.

Percent of Sales Method

Since accounts receivable are created by credit sales, uncollectible accounts can be estimated as a percent of credit sales. If the portion of credit sales to sales is relatively constant, the percent may be applied to total sales.

To illustrate, assume the following data for **ExTone Company** on December 31, 20Y2, before any adjustments:

Balance of Accounts Receivable	\$ 240,000
Balance of Allowance for Doubtful Accounts	3,250 (Cr.)
Total credit sales	3,000,000
Bad debt as a percent of credit sales	$\frac{3}{4}\%$

Business Insight

Allowance Percentages across Companies

The percent of the allowance for doubtful accounts to total accounts receivable will vary across companies and industries. For example, the following percentages were computed from recent annual reports:

Company	Industry	Percent of Allowance for Doubtful Accounts to Total Accounts Receivable
The Coca-Cola Company (KO)	Beverages	11.7%
DuPont (DD)	Chemicals	0.2%
Johnson & Johnson (JNJ)	Pharmaceuticals	1.5%
Nike, Inc. (NKE)	Apparel	0.7%
Union Pacific Corporation (UNP)	Transportation services	0.3%
Wynn Resorts, Limited (WYNN)	Casino gaming	10.6%

Coca-Cola's estimated bad debts stem, in a large part, from collection issues regarding its Venezuelan bottling partner. Wynn Resorts' high percentage is typical for casinos, representing the challenges in collecting gambling debts.

Bad Debt Expense of \$22,500 is estimated as follows:

$$\text{Bad Debt Expense} = \text{Credit Sales} \times \text{Bad Debt as a Percent of Credit Sales}$$

$$\text{Bad Debt Expense} = \$3,000,000 \times \frac{3}{4}\% = \$22,500$$

The adjusting entry for uncollectible accounts on December 31, 20Y2, is as follows:

20Y2					
Dec.	31	Bad Debt Expense	22,500		
		Allowance for Doubtful Accounts		22,500	
		Uncollectible accounts estimate			
		(\$3,000,000 × ¾%).			
		Assets	=	Liabilities	+
		Allowance for Doubtful Accounts			Stockholders' Equity
		<u>22,500</u>			<u>Bad Debt Expense</u>
					<u>22,500</u>

After the adjusting entry is posted to the ledger, Bad Debt Expense will have an adjusted balance of \$22,500. Allowance for Doubtful Accounts will have an adjusted balance of \$25,750 (\$3,250 + \$22,500). Both T accounts follow:

Bad Debt Expense			
Dec. 31, 20Y2	Adjusting entry	<u>22,500</u>	←
Dec. 31, 20Y2	Adjusted balance	<u>22,500</u>	
Allowance for Doubtful Accounts			
Total accounts written off \$26,750 { Jan. 21 6,000 Feb. 2 3,900 ∴ ∴ _____		Jan. 1, 20Y2	Balance 30,000
		Dec. 31, 20Y2	Unadjusted balance <u>3,250</u>
		Dec. 31, 20Y2	Adjusting entry <u>22,500</u> ←
		Dec. 31, 20Y2	Adjusted balance <u>25,750</u>

Under the percent of sales method, the amount of the adjusting entry is the amount estimated for Bad Debt Expense. This estimate is credited to whatever the unadjusted balance is for Allowance for Doubtful Accounts.

To illustrate, assume that in the preceding example the unadjusted balance of Allowance for Doubtful Accounts on December 31, 20Y2, had been a \$2,100 debit balance instead of a \$3,250 credit balance. The adjustment would still have been \$22,500. However, the December 31, 20Y2, ending adjusted balance of Allowance for Doubtful Accounts would have been \$20,400 (\$22,500 – \$2,100).

Note

The estimate based on sales is added to any balance in Allowance for Doubtful Accounts.

Check up Corner 8-1

Percent of Sales Method

At the end of the current year, ARS Industries has the following account balances before making an adjusting entry for uncollectible accounts:

Accounts Receivable	\$800,000 debit
Allowance for Doubtful Accounts	7,500 credit

The company recorded \$3,500,000 of credit sales during the year. The company uses the percent of sales method and estimates that $\frac{1}{2}$ of 1% of credit sales for the year will be uncollectible.

- a. Determine the bad debt expense for the period, and journalize the adjusting entry.
- b. Determine the adjusted balances of Allowance for Doubtful Accounts and Accounts Receivable.
- c. Determine the net realizable value of accounts receivable at the end of the period.

Analysis of Receivables Method

The analysis of receivables method is based on the assumption that the longer an account receivable is outstanding, the less likely that it will be collected. The analysis of receivables method is applied as follows:

Step 1.

The due date of each account receivable is determined.

Step 2.

The number of days each account is past due is determined. This is the number of days between the due date of the account and the date of the analysis.

Step 3.

Each account is placed in an aged class according to its days past due. Typical aged classes include the following:

Days Past Due

Not past due

1–30

31–60

61–90

91–180

181–365

Over 365

Step 4.

The totals for each aged class are determined.

Step 5.

The total for each aged class is multiplied by an estimated percentage of uncollectible accounts for that class.

Step 6.

The estimated total of uncollectible accounts is determined as the sum of the uncollectible accounts for each aged class.

The preceding steps are summarized in an aging schedule, and this overall process is called

aging the receivables (The process of analyzing the accounts receivable and classifying them according to various age groupings, with the due date being the base point for determining age.)

To illustrate, assume that **ExTone Company** uses the analysis of receivables method instead of the percent of sales method. ExTone prepared an aging schedule for its accounts receivable of \$240,000 as of December 31, 20Y2, as shown in **Exhibit 2**.

Exhibit 2

Aging of Receivables Schedule, December 31, 20Y2

	A	B	C	D	E	F	G	H	I
			Not	Days Past Due					
			Past						Over
	Customer	Balance	Due	1–30	31–60	61–90	91–180	181–365	365
Steps 1–3	4 Ashby & Co.	1,500			1,500				
	5 B. T. Barr	6,100					3,500	2,600	
	6 Brock Co.	4,700	4,700						
	7								
	21								
	22 Saxon Woods Co.	600					600		
Step 4	23 Total	240,000	125,000	64,000	13,100	8,900	5,000	10,000	14,000
Step 5	24 Percent uncollectible		2%	5%	10%	20%	30%	50%	80%
Step 6	25 Estimate of uncollectible accounts	26,490	2,500	3,200	1,310	1,780	1,500	5,000	11,200

Assume that ExTone sold merchandise to Saxon Woods Co. on August 29, 20Y2, with terms 2/10, n/30. Thus, the due date ([Step 1](#)) of Saxon Woods' account is September 28, 20Y2, computed as follows:

Credit terms, net	30 days
Less: Aug. 30 and Aug. 31	<u>(2) days</u>
Days in September	<u>28 days</u>

As of December 31, 20Y2, Saxon Woods' account is 94 days past due ([Step 2](#)), computed as follows:

Number of days past due in September	2 days (30 – 28)
Number of days past due in October	31 days
Number of days past due in November	30 days
Number of days past due in December	<u>31 days</u>
Total number of days past due	<u>94 days</u>

[Exhibit 2](#) shows that the \$600 account receivable for Saxon Woods Co. was placed in the 91–180 days past due class ([Step 3](#)).

The total for each of the aged classes is determined ([Step 4](#)). [Exhibit 2](#) shows that \$125,000 of the accounts receivable are not past due, while \$64,000 are 1–30 days past due. ExTone applies a different estimated percentage of uncollectible accounts to the totals of each of the aged classes ([Step 5](#)). As shown in [Exhibit 2](#), the percent is 2% for accounts not past due, while the percent is 80% for accounts over 365 days past due.

The sum of the estimated uncollectible accounts for each aged class ([Step 6](#)) is the estimated uncollectible accounts on December 31, 20Y2. This is the desired adjusted balance for Allowance for Doubtful Accounts. For ExTone, this amount is \$26,490, as shown in [Exhibit 2](#).

Note

The estimate based on receivables is compared to the balance in the allowance account to determine the amount of the adjusting entry.

Comparing the estimate of \$26,490 with the unadjusted balance of the allowance account determines the amount of the adjustment for Bad Debt Expense. For ExTone, the unadjusted balance of the allowance account is a credit balance of \$3,250. The amount to be added to this balance is therefore \$23,240 (\$26,490 – \$3,250). The adjusting entry is as follows:

20Y2			
Dec. 31	Bad Debt Expense	23,240	
	Allowance for Doubtful Accounts		23,240
	Uncollectible accounts estimate		
	(\$26,490 – \$3,250).		

Assets	=	Liabilities	+	Stockholders' Equity
Allowance for Doubtful Accounts				Bad Debt Expense
23,240				23,240

After the preceding adjusting entry is posted to the ledger, Bad Debt Expense will have an adjusted balance of \$23,240. Allowance for Doubtful Accounts will have an adjusted balance of \$26,490, and the net realizable value of the receivables is \$213,510 (\$240,000 – \$26,490). Both T accounts follow:

Bad Debt Expense			
Dec. 31, 20Y2	Adjusting entry	23,240	
Dec. 31, 20Y2	Adjusted balance	<u>23,240</u>	
Allowance for Doubtful Accounts			
	Dec. 31, 20Y2	Unadjusted balance	3,250
	Dec. 31, 20Y2	Adjusting entry	23,240
	Dec. 31, 20Y2	Adjusted balance	<u>26,490</u>

Under the analysis of receivables method, the amount of the adjusting entry is the amount that will yield an adjusted balance for Allowance for Doubtful Accounts equal to that estimated by the aging schedule.

To illustrate, if the unadjusted balance of the allowance account had been a debit balance of \$2,100, the amount of the adjustment would have been \$28,590 (\$26,490 + \$2,100). In this case, Bad Debt Expense would have an adjusted balance of \$28,590. However, the adjusted balance of Allowance for Doubtful Accounts would still have been \$26,490. After the adjusting entry is posted, both T accounts follow:

Bad Debt Expense			
Dec. 31, 20Y2	Adjusting entry	28,590	
Dec. 31, 20Y2	Adjusted balance	<u>28,590</u>	
Allowance for Doubtful Accounts			
Dec. 31, 20Y2	Unadjusted balance	2,100	
	Dec. 31, 20Y2	Adjusting entry	28,590
	Dec. 31, 20Y2	Adjusted balance	<u>26,490</u>

Check up Corner 8-2

Analysis of Receivables Method

At the end of the current year, ARS Industries has the following account balances before making an adjusting entry for uncollectible accounts:

Accounts Receivable \$800,000 debit

Allowance for Doubtful
Accounts

7,500 credit

The company recorded \$3,500,000 of credit sales during the year. An aging of the company's accounts receivable on December 31 and a historical analysis of the percentage of uncollectible accounts in each age class follow:

Age Class	Balance	Percent Uncollectible
Not past due	\$658,000	1%
1–30 days	69,000	6
31–90 days	47,000	14
Over 90 days	26,000	20
	<u>\$800,000</u>	

The company estimates the allowance for doubtful accounts based on the analysis of receivables method.

- Determine the bad debt expense for the period, and journalize the adjusting entry.
- Determine the adjusted balances of Allowance for Doubtful Accounts and Accounts Receivable.
- Determine the net realizable value of accounts receivable at the end of the period.

Comparing Estimation Methods

Both the percent of sales and analysis of receivables methods estimate uncollectible accounts. However, each method has a slightly different focus and financial statement emphasis.

Under the percent of sales method, Bad Debt Expense is the focus of the estimation process. The percent of sales method places more emphasis on matching revenues and expenses and, thus, emphasizes the income statement. That is, the amount of the adjusting entry is based on the estimate of Bad Debt Expense for the period. Allowance for Doubtful Accounts is then credited for this amount.

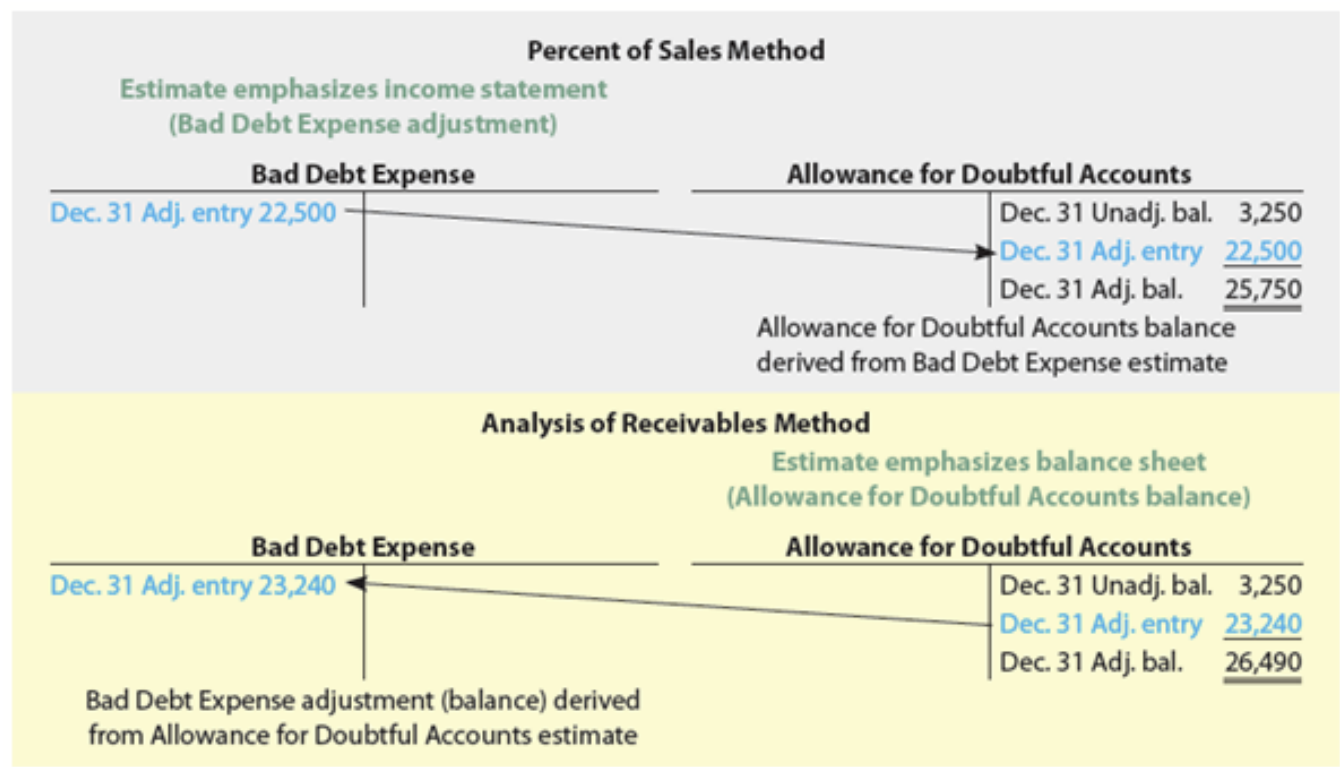
Under the analysis of receivables method, Allowance for Doubtful Accounts is the focus of

the estimation process. The analysis of receivables method places more emphasis on the net realizable value of the receivables and, thus, emphasizes the balance sheet. That is, the amount of the adjusting entry is the amount that will yield an adjusted balance for Allowance for Doubtful Accounts equal to that estimated by the aging schedule. Bad Debt Expense is then debited for this amount.

Exhibit 3 summarizes these differences between the percent of sales and the analysis of receivables methods. Exhibit 3 also shows the results of the ExTone Company illustration for the percent of sales and analysis of receivables methods. The amounts shown in Exhibit 3 assume an unadjusted credit balance of \$3,250 for Allowance for Doubtful Accounts. While the methods normally yield different amounts for any one period, over several periods the amounts should be similar.

Exhibit 3

Difference between Estimation Methods



Using Data Analytics

Collectability of Credit Sales

Most retail businesses accept only cash or credit card payments from customers. As a result, these companies do not have to assess the creditworthiness of their customers or the collectability of their credit sales (accounts receivable). For these businesses, “cash and carry” is an efficient way of doing business.

Other types of businesses, such as suppliers of merchandise to retail businesses, frequently ship merchandise on credit. For example, suppliers such as The Procter

& Gamble Company (PG) sell merchandise to a variety of different types of retail stores, from local convenience stores to large retail chains such as **Costco (COST)**, **Target (TGT)**, and **Walmart (WMT)**.

Companies like Procter & Gamble can use data analytics to assess the collectability and timeliness of payments from different classes of customers. For example, large retail chains such as Costco would normally pay their receivables but might take longer to pay. In contrast, credit sales to small, locally owned stores might be more likely to be uncollectible. By using data analytics, a retail supplier might implement different credit procedures and credit terms for each class of customer.

See **TIF 8-6** for a homework assignment using data analytics.

Chapter 8: Receivables: 8-4b Estimating Uncollectibles

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